

2.3 - Differentiation Rules

Goals: State and apply Derivative Rules

Before Class Videos: 2.3.1

Theorem (Power Rule):

- $\frac{d}{dx}(c) = 0$ where c is a constant.
- $\frac{d}{dx}(x^n) = nx^{n-1}$ where $n > 0$ is an integer.

Theorem:

If $f(x)$ and $g(x)$ are differentiable at $x = a$, then

1. **Sum/Difference Rule:**

$$\frac{d}{dx}[f(x) \pm g(x)] = \frac{d}{dx}f(x) \pm \frac{d}{dx}g(x)$$

2. **Constant Multiple Rule:**

$$\frac{d}{dx}[cf(x)] = c\frac{d}{dx}f(x)$$

Example 1: Compute the derivative of $f(x) = 3x^4 - 5\sqrt{x} + 6x^{3/2} - \frac{7}{x^2} + 1$.

Proof of Theorem:

Common Derivatives:

Example 2: Explore the graphs of the following functions to discover their derivatives.

- $\frac{d}{dx}(e^x) =$

- $\frac{d}{dx} \ln(x) =$ for $x > 0$

- $\frac{d}{dx} \sin(x) =$

- $\frac{d}{dx} \cos(x) =$

Higher Derivatives: The derivative $f'(x)$ is itself a function we can take the derivative of. We call this the second derivative and might denote it by

$$f''(x) \quad \text{or} \quad \frac{d^2}{dx^2} f(x).$$

Example 3: Compute $f''(x)$ when $f(x) = e^x + \ln(x)$.

Example 4: If $f(x) = \sin(x)$ compute

- $f'(x) =$

- $f''(x) =$

- $f'''(x) =$

- $f^{(4)}(x) =$

- $f^{(2026)}(x) =$

Example 5: If $d(t)$ represents the position of a particle as a function of time, interpret $d'(t)$ and $d''(t)$.

Example 6: If $v(t)$ represents Trefor's total YouTube views after t days, interpret $v'(t)$ and $v''(t)$.

Definition:

The Point-Slope formula for the equation of the straight line through (x_0, y_0) with slope m is

$$y = y_0 + m(x - x_0)$$

Example 7: Find the equation of tangent line to a differentiable function $f(x)$ at the point x_0 .

Example 8: Find the equation of the tangent line to the function $f(x) = 2x^2 + 1$ at the point $(1, 3)$.